

CBSE 2027 Chemistry — Predicted Paper (Set 7 of 10)

Based on the highly repeated questions from previous years' board exams (PYQs), official CBSE directives, and the most critical NCERT concepts, here is Set 7 of the predicted board paper. All questions are completely new and distinct from the previous sets, targeting heavily tested formulas, name reactions, and analytical reasoning.

SECTION A: Multiple Choice & Assertion-Reason Questions (1 Mark Each x 16 = 16 Marks)

Q1. The Van't Hoff factor (i) for a dilute aqueous solution of Aluminium Sulphate, $Al_2(SO_4)_3$, assuming complete dissociation, is: (a) 2 (b) 3 (c) 4 (d) 5 * **Chapter:** Solutions | **Confidence:** Very High

Q2. During the electrolysis of aqueous Sodium Chloride ($NaCl$) solution using inert electrodes, the gas liberated at the cathode is: (a) Cl_2 (b) O_2 (c) H_2 (d) Na vapour * **Chapter:** Electrochemistry | **Confidence:** High

Q3. For a chemical reaction, the rate law is given as $\text{Rate} = k[A]^1[B]^2$. If the volume of the reaction vessel is suddenly reduced to one-third of its original volume, the rate of the reaction will become: (a) 9 times (b) 27 times (c) 1/9 times (d) 3 times * **Chapter:** Chemical Kinetics | **Confidence:** High

Q4. Which of the following lanthanoids exhibits a stable +4 oxidation state due to the formation of a noble gas electronic configuration? (a) Europium (Eu) (b) Ytterbium (Yb) (c) Cerium (Ce) (d) Lutetium (Lu) * **Chapter:** d- and f-Block Elements | **Confidence:** Very High

Q5. The denticity of the $EDTA^{4-}$ (Ethylenediaminetetraacetate) ligand is: (a) 2 (bidentate) (b) 4 (tetradentate) (c) 6 (hexadentate) (d) 1 (unidentate) * **Chapter:** Coordination Compounds | **Confidence:** High

Q6. The most preferred reagent for converting a primary alcohol into an alkyl chloride, because the by-products formed are escapable gases, is: (a) PCl_3 (b) PCl_5 (c) $HCl/ZnCl_2$ (d) $SOCl_2$ * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** Very High

Q7. What is the correct decreasing order of acidic strength among the following alcohols? (a) Primary > Secondary > Tertiary (b) Tertiary > Secondary > Primary (c) Secondary > Primary > Tertiary (d) Primary > Tertiary > Secondary * **Chapter:** Alcohols, Phenols and Ethers | **Confidence:** High

Q8. Which of the following carboxylic acids will **not** undergo the Hell-Volhard-Zelinsky (HVZ) reaction? (a) Propanoic acid (b) Ethanoic acid (c) 2-Methylpropanoic acid (d) Benzoic acid * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** Very High

Q9. When a primary aliphatic or aromatic amine is heated with chloroform ($CHCl_3$) and ethanolic potassium hydroxide (KOH), it forms a highly foul-smelling compound known as:

(a) An amide (b) An isocyanide (c) A diazonium salt (d) A secondary amine * **Chapter:** Amines | **Confidence:** Very High

Q10. Which of the following is an optically inactive α -amino acid? (a) Alanine (b) Valine (c) Glycine (d) Leucine * **Chapter:** Biomolecules | **Confidence:** High

Q11. The reaction converting an alkyl halide into an alkene by heating it with alcoholic KOH is categorized as: (a) Nucleophilic substitution (b) Electrophilic addition (c) β -Elimination (d) Free radical substitution * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** High

Q12. According to the spectrochemical series, which of the following is the strongest field ligand? (a) SCN^- (b) H_2O (c) CO (d) NH_3 * **Chapter:** Coordination Compounds | **Confidence:** Very High

Directions for Q13 to Q16: Two statements are given—one labeled Assertion (A) and the other Reason (R). Select the correct answer from the codes (a), (b), (c), and (d). (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

Q13. Assertion (A): The half-life period of a first-order chemical reaction is completely independent of the initial concentration of the reactant. **Reason (R):** The half-life of a first-order reaction is given by the formula $t_{1/2} = \frac{0.693}{k}$. * **Chapter:** Chemical Kinetics | **Confidence:** Very High

Q14. Assertion (A): The rate of a chemical reaction continuously decreases as the reaction proceeds. **Reason (R):** The concentration of the reacting species decreases continuously over time. * **Chapter:** Chemical Kinetics | **Confidence:** High

Q15. Assertion (A): Copper (Cu , $Z=29$) is considered a transition metal despite having a completely filled d-subshell ($3d^{10}$) in its ground state. **Reason (R):** Copper exhibits a +2 oxidation state (Cu^{2+}) in its common compounds, which possesses an incompletely filled d-subshell ($3d^9$). * **Chapter:** d- and f-Block Elements | **Confidence:** Very High

Q16. Assertion (A): The electrophilic substitution of benzoic acid (e.g., nitration) produces a meta-substituted product. **Reason (R):** The carboxyl group ($-COOH$) is an electron-withdrawing group and acts as a meta-directing group by deactivating the ortho and para positions via resonance. * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** Very High

SECTION B: Very Short Answer Questions (2 Marks Each x 5 = 10 Marks)

Q17. Calculate the mole fraction of the solute in a exactly 1.0 molal aqueous solution. * **Chapter:** Solutions | **Confidence:** High

Q18. Draw the structures of the geometrical isomers (*cis* and *trans*) of the complex ion $[Co(NH_3)_4Cl_2]^+$. * **Chapter:** Coordination Compounds | **Confidence:** High

Q19. Define the term *limiting molar conductivity* (Λ_m°). Why is it difficult to determine the limiting molar conductivity of a weak electrolyte experimentally by extrapolating the graph of

Λ_m vs $\sqrt{c}$? * **Chapter:** Electrochemistry | **Confidence:** Very High

Q20. The rate constant for a reaction possesses the units $L^2 \cdot mol^{-2} \cdot s^{-1}$. What is the overall order of the reaction? Briefly define what a *pseudo-first-order reaction* is. * **Chapter:** Chemical Kinetics | **Confidence:** High

Q21. Give simple chemical tests with observable outcomes to distinguish between the following pairs: (i) 1-Propanol and 2-Propanol (ii) Phenol and Ethanol * **Chapter:** Alcohols, Phenols and Ethers | **Confidence:** Very High

SECTION C: Short Answer Questions (3 Marks Each x 7 = 21 Marks)

Q22. Answer the following reasoning questions regarding amines: (i) Why is aniline a weaker base than ammonia? (ii) Why is Gabriel phthalimide synthesis preferred for synthesizing primary aliphatic amines but cannot be used for the preparation of primary aromatic amines? * **Chapter:** Amines | **Confidence:** Very High

Q23. An alkyl halide CH_3CH_2Br (Bromoethane) reacts with ethanolic KCN to give compound **A**. Compound **A** on reduction with $LiAlH_4$ yields compound **B**. Compound **B** on treatment with nitrous acid (HNO_2) at $0-5^\circ C$ produces compound **C**. Identify the structures and IUPAC names of compounds **A**, **B**, and **C**. * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** High

Q24. Write the Nernst equation and calculate the electromotive force (EMF) of the following cell at 298 K: $Cu(s)|Cu^{2+}(0.01 M)||Ag^+(0.001 M)|Ag(s)$ Given: $E^\circ_{Cu^{2+}/Cu} = +0.34 V$ and $E^\circ_{Ag^+/Ag} = +0.80 V$. ($\log 10 = 1$) * **Chapter:** Electrochemistry | **Confidence:** Very High

Q25. Answer the following biochemical questions: (i) What are essential amino acids? Give one example. (ii) Differentiate between fibrous proteins and globular proteins based on their shape and solubility. (iii) What is a zwitterion? * **Chapter:** Biomolecules | **Confidence:** High

Q26. Complete the following essential organic reactions by writing the structures of the main products: (i) Toluene reacts with Chromyl Chloride (CrO_2Cl_2) in CS_2 followed by hydrolysis. (ii) Benzene reacts with Carbon Monoxide (CO) and HCl in the presence of anhydrous $AlCl_3$. (iii) Propanone reacts with Iodine (I_2) and aqueous $NaOH$ under heating. * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** Very High

Q27. Account for the following transition metal properties: (i) Scandium (III) salts are colorless in an aqueous solution. (ii) Interstitial compounds formed by transition metals are generally very hard. (iii) The highest oxidation states of transition metals are found only in their fluorides and oxides. * **Chapter:** d- and f-Block Elements | **Confidence:** Very High

Q28. The plot of $\ln k$ versus $\frac{1}{T}$ for a chemical reaction yields a straight line with a slope of $-5000 K$. Calculate the activation energy (E_a) for the reaction. ($R = 8.314 J \cdot K^{-1} \cdot mol^{-1}$). * **Chapter:** Chemical Kinetics | **Confidence:** High

SECTION D: Case Study-Based Questions (4 Marks Each x 2 = 8 Marks)

Q29. Case Study 1: Osmosis and Osmotic Pressure When a semipermeable membrane separates a solution from a pure solvent, solvent molecules pass through the membrane into the solution. This spontaneous flow of solvent molecules through a semipermeable membrane from a pure solvent to a solution (or from a dilute solution to a concentrated solution) is called osmosis. The external pressure that must be applied to the solution side to just stop the flow of solvent is called osmotic pressure (π). Two solutions having the same osmotic pressure at a given temperature are called isotonic solutions. If a cell is placed in a hypertonic solution, water flows out of the cell, causing it to shrink. (i) What happens to a human red blood cell (RBC) when it is placed in a hypotonic solution? (1 Mark) (ii) State the mathematical formula that relates osmotic pressure (π) to the molarity (C) of the solution. (1 Mark) (iii) A 5% (by mass/volume) aqueous solution of cane sugar (Molar mass = 342 g/mol) is isotonic with a 1% (by mass/volume) aqueous solution of an unknown solute X. Assuming both solutions behave ideally, calculate the molar mass of the unknown solute X. (2 Marks) *

Chapter: Solutions | **Confidence:** Very High

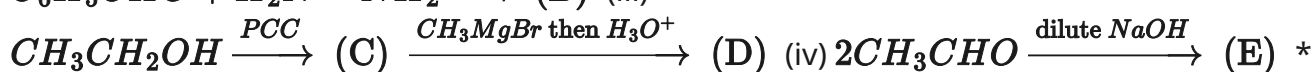
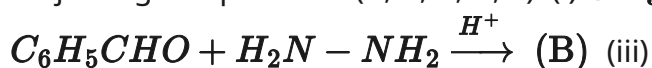
Q30. Case Study 2: Diazonium Salts in Organic Synthesis Benzene diazonium chloride ($C_6H_5N_2^+Cl^-$) is an incredibly important synthetic intermediate in organic chemistry. It is prepared by reacting aniline with nitrous acid (generated *in situ* from $NaNO_2$ and HCl) at cold temperatures (273-278 K), a process known as diazotization. The diazonium group is an excellent leaving group because it departs as highly stable nitrogen gas (N_2). Thus, it can be easily replaced by various nucleophiles such as halide ions, cyanide, or hydroxyl groups, facilitating the synthesis of numerous substituted benzene derivatives that are difficult to prepare by direct electrophilic substitution. (i) Why is the temperature strictly maintained between 0°C and 5°C (273-278 K) during the preparation of benzene diazonium chloride? (1 Mark) (ii) What chemical product is formed when benzene diazonium chloride is warmed directly with water (H_2O)? (1 Mark) (iii) Write the complete chemical equations for the conversion of benzene diazonium chloride into: a) Chlorobenzene (Sandmeyer Reaction) b) Benzene (using Hypophosphorous acid, H_3PO_2) (2 Marks) * **Chapter:** Amines | **Confidence:** High

SECTION E: Long Answer Questions (5 Marks Each x 3 = 15 Marks)

Q31. (A) Physical Chemistry: Solutions & Colligative Properties (i) State Raoult's law for a solution containing non-volatile solutes. (1 Mark) (ii) Explain with a suitable diagram why the boiling point of a solvent elevates upon the addition of a non-volatile solute. (2 Marks) (iii) Calculate the boiling point of an aqueous solution containing 15.0 g of urea (NH_2CONH_2 , Molar mass = 60 g/mol) dissolved in 250 g of water. (Given: K_b for water = $0.52 K \cdot kg \cdot mol^{-1}$, Boiling point of pure water = 373.15 K). (2 Marks) * **OR (B)** (i) Define the term *azeotrope*. Give one example of a minimum-boiling azeotrope. (2 Marks) (ii) State Henry's law. Mention any two critical real-life applications of Henry's law. (2 Marks) (iii) Why

does the solubility of a gas in a liquid generally decrease with an increase in temperature? (1 Mark) * **Chapter:** Solutions | **Confidence:** Very High

Q32. (A) Organic Chemistry: Multi-step Synthesis and Reasoning An aromatic organic compound **(A)** with the molecular formula C_7H_6O gives a silver mirror with Tollens' reagent. When compound **(A)** is treated with concentrated aqueous KOH (50%), it undergoes a disproportionation reaction to form two compounds, **(B)** and **(C)**. Compound **(B)** is an alcohol that, upon oxidation, regenerates compound **(A)**. Compound **(C)** is the potassium salt of a carboxylic acid, which upon acidification yields compound **(D)**. Compound **(D)** produces brisk effervescence when treated with an aqueous solution of sodium bicarbonate ($NaHCO_3$). (i) Identify the structures and IUPAC names of compounds **(A)**, **(B)**, **(C)**, and **(D)**. (4 Marks) (ii) Name the specific chemical reaction that converts compound **(A)** into compounds **(B)** and **(C)**. (1 Mark) * **OR (B)** Complete the following reactions by providing the structures of the missing major organic products (A, B, C, D, E): (i) $CH_3CH_2COOH \xrightarrow{Cl_2 / \text{Red P}}$ (A) (ii)



Chapter: Aldehydes, Ketones and Carboxylic Acids | **Confidence:** Very High

Q33. (A) Electrochemistry: Faraday's Laws & Batteries (i) State Michael Faraday's Second Law of Electrolysis. (1 Mark) (ii) A steady direct current of 3.0 Amperes is passed through a solution of Silver Nitrate ($AgNO_3$) for exactly 2 hours. Calculate the mass of silver metal deposited at the cathode. (Given: Molar mass of Ag = 108 g/mol , $1F = 96500 \text{ C/mol}$). (2 Marks) (iii) Briefly describe the construction and the charging reactions of a Lead Storage Battery. What acts as the anode, cathode, and electrolyte? (2 Marks) * **OR (B)** (i) Explain how the molar conductivity (Λ_m) of a strong electrolyte varies with a decrease in concentration (dilution). Draw the expected theoretical plot of Λ_m vs $\sqrt{c}$. (2 Marks) (ii) Given the standard reduction potentials: $E^\circ_{Zn^{2+}/Zn} = -0.76 \text{ V}$, $E^\circ_{Cr^{3+}/Cr} = -0.74 \text{ V}$, $E^\circ_{Fe^{2+}/Fe} = -0.44 \text{ V}$. Predict which among these metals (Zn , Cr , or Fe) is the strongest reducing agent and justify your choice. (1 Mark) (iii) Write the overall cell reaction and calculate the standard Gibbs free energy change (ΔG°) for a standard cell constructed from Zn/Zn^{2+} and Fe/Fe^{2+} half-cells. ($1F = 96500 \text{ C/mol}$). (2 Marks) * **Chapter:** Electrochemistry | **Confidence:** Very High

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.